

A 61-year-old man presents with a large haematemesis. He has known alcoholic liver disease and sedation is requested for an urgent gastroscopy.

A. Outline your peri-procedural concerns (60%).

Towards the end of the procedure, the systolic blood pressure drops from 140 mmHg to 80 mmHg systolic, and oxygen saturation drops from 95% to 88%.

B. Describe your subsequent management (40%).

A.

This is a challenging case involving:

- Elderly patient with
- Possibly decompensated liver disease
- Haematemesis, bleeding risks
- High aspiration risk
- Presenting for emergency procedure under sedation (unsecured airway)
- In remote location (endoscopy room)

Peri-procedural concerns include:

Pre-procedure

Haematemesis

- Difficult to quantify blood loss due to swallowed blood
- Serum Hb is a lagging indicator.
- Will need valid GXM, standby blood products

Possible decompensated liver disease, in the background of alcoholic liver disease

- Haematemesis may be made worse with underlying thrombocytopaenia and coagulopathy secondary to chronic liver disease.
- Fluid resuscitation more challenging.
- If patient has alcoholic/ liver cardiomyopathy/ hyperdynamic circulation.
- Compounded by third spacing (pleural effusions/ ascites) secondary to hypoalbuminemia.
- Possibly worsening AKI secondary to dehydration, background of possible hepatorenal syndrome.
- Increased risks of respiratory complications from hepatopulmonary syndrome, pulmonary hypertension, orthodeoxia, and diaphragmatic splinting from ascites.
- Hepatic encephalopathy may make patient cooperation and assessment difficult.

Alcohol intoxication/ withdrawal/ delirium tremens

- Unknown duration regarding last alcohol intake
- Acute intoxication affects patient cooperation. Caution with oversedation.
- Risks of alcohol withdrawal/ delirium tremens which is potentially life threatening.
- Thiamine and Vit B12 should be administered pre-procedure.

Urgency of procedure vs time for medical optimisation

- Patient will likely require fluid (and possibly blood product) resuscitation, possibly correction of electrolytes, metabolic acidosis, and coagulopathy which may worsen bleeding.
- This has to be balanced against the urgency of procedure, especially if haematemesis is massive and still ongoing.

Intra-procedure

Poor candidate for sedation due to

- Possible challenges with cooperation
- Importance to secure airway due to aspiration risks and limited airway access once gastroscopy has started
- Sedation not contraindicated but will require patient cooperation and deliberated whether haematemesis is still ongoing.
- This has to be balanced against the risks of increased haemodynamic instability with GA.

Limited access to airway/ shared airway

- Any suggestion of difficult airway/ risks of airway obstruction would necessitate securing airway
- Careful taping/ positioning of ETT with close monitoring to kinks/ dislodgement/ malposition during procedure.

High risk of aspiration

- Due to haematemesis
- Impaired ability to protect airway under sedation
- Worse if patient has concomitant hepatic encephalopathy, acute intoxication or delirium tremens.
- Decompression of stomach at the end of procedure prior to extubation is necessary.

Management of bleeding and haemodynamic instability

- 2x Large bore IV access connected to warming device for blood administration. Consider rapid fluid infuser.
- Appropriate haemodynamic monitors would include insertion of IA line (and possibly cardiac output monitor such as Flotrac) on top of standard ASA monitors.
- BIS monitor to allow for titration of GA and minimise excessive depth of anaesthesia.
- This facilitates POCT ABG, Hb and ROTEM check.
- Maintain Hb 8-10 g/dL. Consider TXA administration.
- Correction of coagulopathy will need to be titrated based on both ROTEM as well as clinical haemostasis.
- Avoid lethal triad. Metabolic acidosis (to correct based on ABG) and hypothermia (temp monitoring, active warming devices) should be minimised.
- NSAIDs/ COX-II inhibitors should be avoided.

Altered drug metabolism secondary to

- Elderly patient physiology
- Chronic liver disease
- AKI, or presence of hepatorenal syndrome
- These will necessitate dose adjustment anaesthetic agents (Propofol), opioids, other analgesics such as paracetamol, aminosteroid NMBA agents.

Remote location

- Limited access to prompt help
- Endoscope room not optimised for administration of GA, resuscitation.
- Discussion with proceduralist if OT is more appropriate location for gastroscope.

Post-procedure

Recovery from sedation/ GA may be prolonged

- Due to impaired physiology/ drug clearance
- Require longer duration of monitoring in recovery

Need for close haemodynamic monitoring

- HD more appropriate post-procedure disposition compared to GW

B.

Assuming that patient underwent sedation for procedure (as mentioned the stem),

- This is a concerning situation which should be immediately attended to.
- I will inform the proceduralist and ask if there is any worsening bleeding/ missed spots for bleeding.
- Call my senior and additional nursing help.
- And proceed to simultaneously assess and treat patient based on ABCs.

Important causes for hypotension and desaturation include:

- Massive haematemesis leading to hypovolaemic shock and poor peripheral perfusion.
- Aspiration leading to desaturation
- Decompensated liver disease with congestive cardiomyopathy/ fluid overload, with desaturation from atelectasis/ pulmonary congestion/ hepatopulmonary syndrome.
- Anaphylaxis with distributive shock and bronchospasm
- Tension pneumothorax with obstructive shock

Airway

- Patient likely in lateral position.
- Head tilt, chin lift, jaw thrust, nasal airway.
- Low threshold for intubation if unable to ascertain airway patency, or if ongoing haematemesis.
- Patient should be turned supine, head down. To proceed with SALAD – Suction Assisted Laryngoscopy Airway Decontamination.

Breathing

- Administered FiO₂ 1.0 via HFNC (if in-situ) or bag mask ventilation using anaesthesia breathing circuit
- Check for adequacy of respiratory effort, with support ventilation as required.
- Auscultate bilateral lungs for unequal breath sounds (pneumothorax), wheeze/ rhonchi (bronchospasm), crepitations (aspiration, fluid overload, atelectasis).
- Salbutamol puffs post-intubation as required.

Circulation

- Check pulse.
- Check rest of vital signs + ECG for signs of ischemia/ arrhythmias.
- Run fluids fast. Trendelenburg position. Insert another large bore IV cannula if required.
- Stop sedation.
- Call for blood products if bleeding is suspected as cause of hypotension.
- Vasopressors/ adrenaline as required.

Examine for angioedema, rash suggestive of anaphylaxis.

Send off POCT ABG, Hb, ROTEM.

POCUS heart and lungs to ascertain cause for hypotension and desaturation.

Monitor for improvement after treatment, and re-assess patient's ABC if situation does not improve/ worsens.

Subsequent treatment will need to be targeted to underlying cause.

Discussion with proceduralist regarding continuing procedure vs sending to ICU intubated for stabilisation.